Lecture #2

Regression Analysis

Topic: Regression equations (X on Y & Y on X)

Probability and Statistics for engineers
&
Scientists

by

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Regression line of x on y

$$x - \bar{x} = b_{xy}\,(y - \bar{y})$$

$$b_{xy} = \frac{n\sum xy - \sum x \sum y}{n\sum y^2 - (\sum y)^2}$$

$y = independant\ variable$

$x = dependant\ variable$

$use\ to\ calculate\ x\ for\ given\ y$

$slope\ of\ line = b_{xy}$

Regression line of y on x

$$y - \bar{y} = b_{yx}\,(x - \bar{x})$$

$$b_{yx} = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$y = dependant\ variable$

$x = independant\ variable$

$use\ to\ calculate\ y\ for\ given\ x$

$slope\ of\ line = b_{yx}$

Question: The data about expenditure (X) & income (Y) (in thousand rupee) of nine families is given in following table.

Expenditure (x)	Income (y)
9	15
8	16
7	14
6	13
5	11
4	12
3	10
2	8
1	9

1. Find both regression lines/least square lines/regression equation for the following data.
2. Draw a scatter diagram.
3. Estimate expenditure if income is 90.
4. Sketch the regression line on same diagram
5. Show that sum of residuals/ errors is zero
6. Find the correlation coefficient and interpret it.

Correlation and Regression

Question: Find both regression lines for the following data

x	y
9	15
8	16
7	14
6	13
5	11
4	12
3	10
2	8
1	9

Regression line of x on y

$$x - \bar{x} = b_{xy}\,(y - \bar{y})$$

$$b_{xy} = \frac{n\sum xy - \sum x \sum y}{n\sum y^2 - (\sum y)^2}$$

Regression line of y on x

$$y - \bar{y} = b_{yx}\,(x - \bar{x})$$

$$b_{yx} = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$\sum xy = ?$

$\sum x = ?$

$\sum y =$

$\sum x^2 =$

$\sum y^2 =$

$\bar{x} =$

$\bar{y} =$

Find coefficient of correlation of the following data

$\sum xy = 597$

$\sum x = 45$

$\sum y = 108$

$\sum x^2 = 285$

$\sum y^2 = 1356$

x	y	x^2	y^2	xy
9	15	81	225	135
8	16	64	256	128
7	14	49	196	98
6	13	36	169	78
5	11	25	121	55
4	12	16	144	48
3	10	9	100	30
2	8	4	64	16
1	9	1	81	9
$\sum x = 45$	$\sum y = 108$	$\sum x^2 = 285$	$\sum y^2 = 1356$	$\sum xy = 597$

$$\bar{x} = \frac{\sum x}{n} = \frac{45}{9} = 5$$

$$b_{xy} = \frac{n\sum xy - \sum x \sum y}{n\sum y^2 - (\sum y)^2}$$

$$b_{xy} = \frac{9(597)-(45)(108)}{9(1356)- \ (108)^2} = 0.95$$

$$\bar{y} = \frac{\sum y}{n} = \frac{108}{9} = 12$$

$$b_{yx} = \frac{n\sum xy - \sum x \sum y}{n\sum x^2 - (\sum x)^2}$$

$$b_{yx} = \frac{9(597)-(45)(108)}{9(285)- \ (45)^2} = 0.95$$

Regression line of x on y

$$x - \bar{x} = b_{xy}\,(y - \bar{y})$$

$$x - 5 = 0.95\,(y - 12)$$

$$x - 5 = 0.95y - 11.4$$

$$\boldsymbol{x = 0.95y - 6.4}$$

Regression line of y on x

$$y - \bar{y} = b_{yx}\,(x - \bar{x})$$

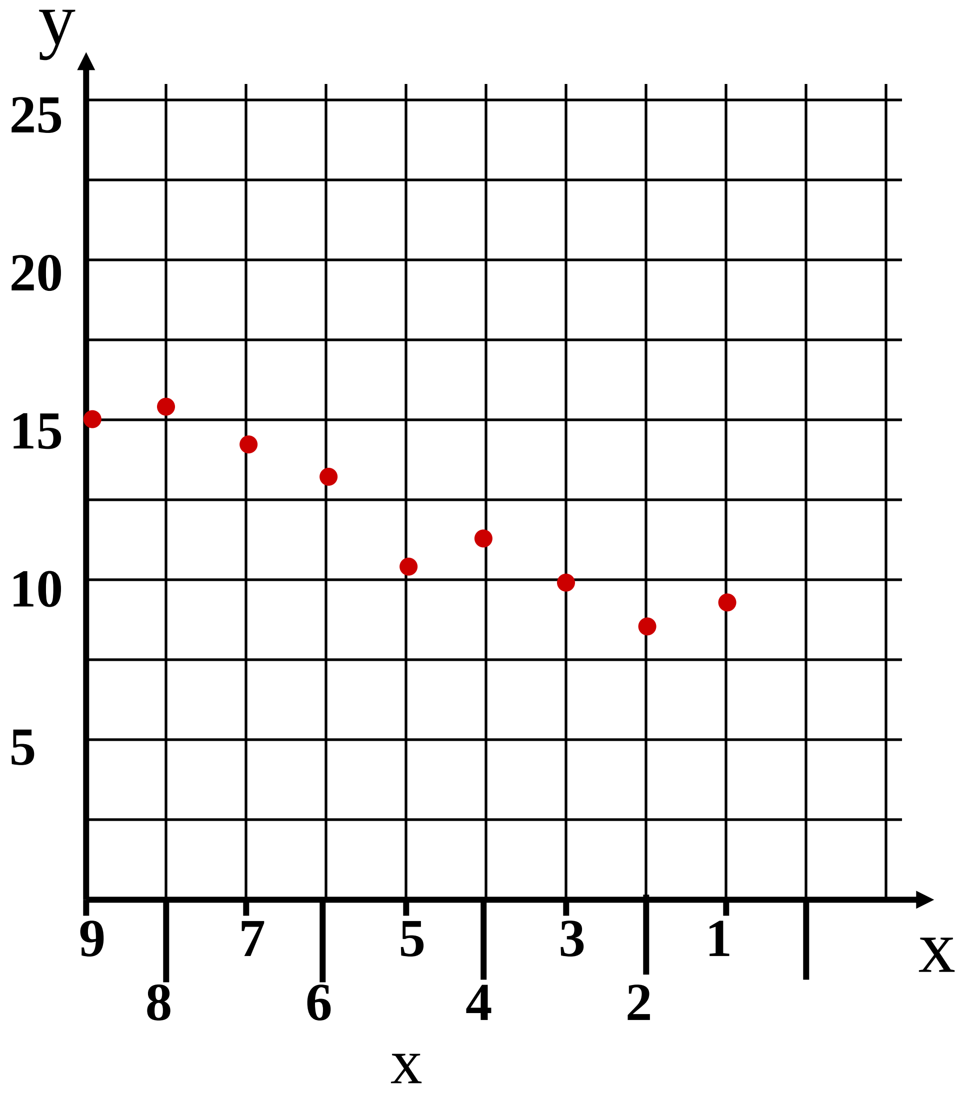
$$y - 12 = 0.95\,(x - 5)$$

$$y - 12 = 0.95x - 4.75$$

$$\boldsymbol{y = 0.95x + 7.25}$$

Objective - To plot data points in the coordinate plane and interpret scatter plots.

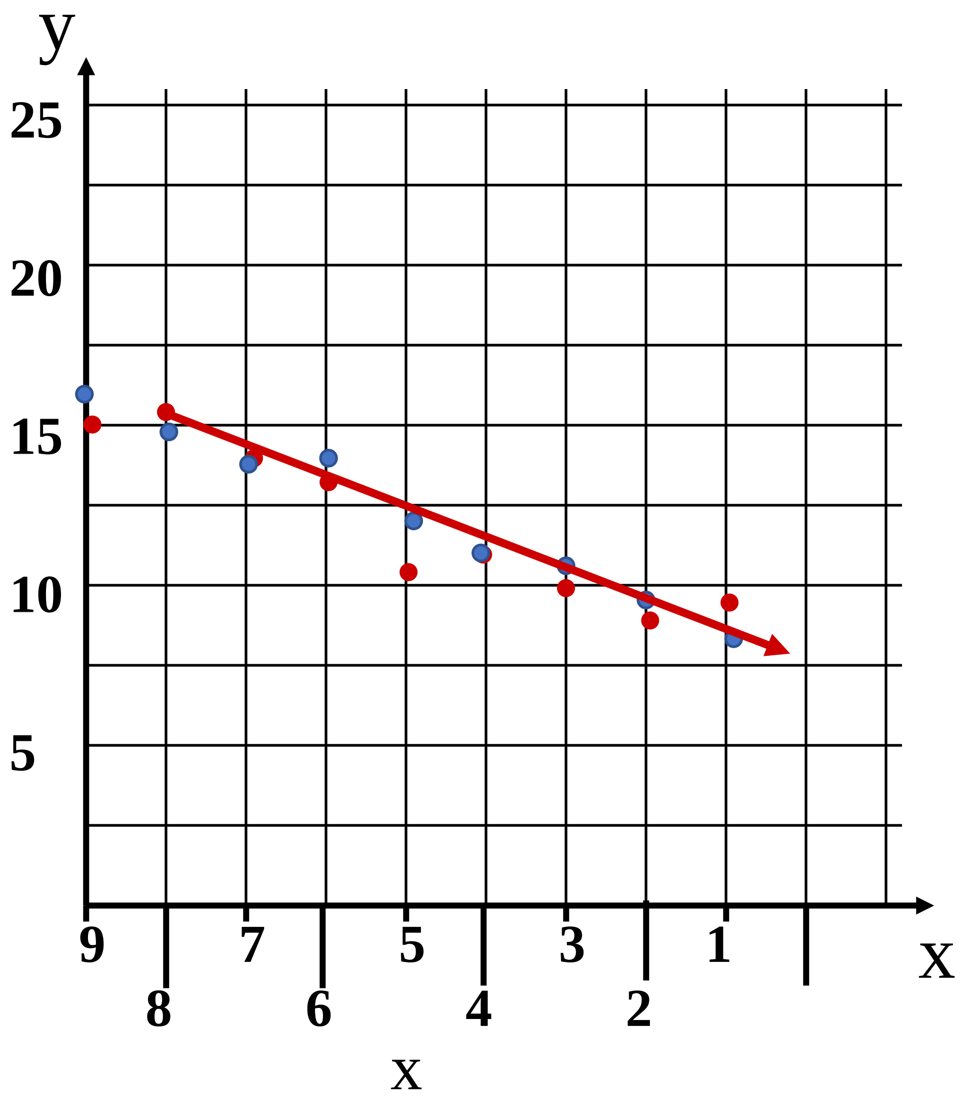
x	y
9	15
8	16
7	14
6	13
5	11
4	12
3	10
2	8
1	9



Objective - To plot data points in the coordinate plane and interpret scatter plots.

Estimation of Y on X

x	$\hat{y} = 0.95x + 7.25$
9	15.8
8	14.85
7	13.9
6	12.95
5	12
4	11.05
3	10.1
2	9.15
1	8.2



Find error, and show that sum of residuals is zero

x	y
9	15
8	16
7	14
6	13
5	11
4	12
3	10
2	8
1	9

x	$\hat{y} = 0.95x + 7.25$
9	15.8
8	14.85
7	13.9
6	12.95
5	12
4	11.05
3	10.1
2	9.15
1	8.2

x	$e = y - \hat{y}$
9	-0.8
8	1.15
7	0.1
6	0.05
5	-1
4	0.95
3	-0.1
2	-1.15
1	0.8

$$\sum e = 0$$

Correlation Coefficient

$$\boldsymbol{r} = \frac{\boldsymbol{n}\sum \boldsymbol{xy} - \sum \boldsymbol{x}\sum \boldsymbol{y}}{\sqrt{\boldsymbol{n}\sum \boldsymbol{x}^2 - (\sum \boldsymbol{x})^2}\ \sqrt{\boldsymbol{n}\sum \boldsymbol{y}^2 - (\sum \boldsymbol{y})^2}}$$

$$r = \frac{9(597)-(45)(108)}{\sqrt{9\,(285)-(45)^2}\ \sqrt{9(1356)-(108)^2}} = 0.95$$

Strong Correlation